

Evaluation of serological tests for the diagnosis of *Neospora caninum* infection in dogs: Optimization of cut off titers and inhibition studies of cross-reactivity with *Toxoplasma gondii*

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Abstract

Diagnosis of *Neospora caninum* infection in dogs is based on serological assays such as the indirect fluorescent antibody test (IFAT) and enzyme-linked immunosorbent assays (ELISA). This study evaluated two serological tests (IFAT and ELISA) for the detection of IgG antibodies to *N. caninum* in 300 serum samples of dogs through the optimization of cut off titers by using the two-graph receiver-operating characteristic (TG–ROC) curve. In addition, the identification of major cross-reactive antigens with *Toxoplasma gondii* was investigated by inhibition ELISA and immunoblotting (IB) assays. IFAT and ELISA results showed 74% agreement, with a good negative concordance ($P_{\text{neg}} = 0.83$), but a poor positive concordance ($P_{\text{pos}} = 0.42$). The great majority (86%) of sera with positive concordant results (IFAT+/ELISA+) recognized at least two out of three *N. caninum* immunodominant antigens, particularly the 29–32 and 35–37 kDa bands. Optimization of cut off titers in IFAT and ELISA was performed considering the reactivity to at least two out of three *N. caninum* immunodominant antigens as infection markers, obtaining a titer of 50 for IFAT and 200 for ELISA. Seropositivity to *N. caninum* was significantly associated with *T. gondii*-seropositive samples, particularly in ELISA (55.4%). Inhibition ELISA curves for *N. caninum* showed a partial heterologous inhibition, indicating some degree of cross-reactivity between *N. caninum* and *T. gondii* antigens. Inhibition IB assays showed a moderate heterologous inhibition for *N. caninum* antigens above 45–50 kDa. These results indicate that ELISA should be used critically when crude tachyzoite antigen preparations are employed, due to possible cross-reactivity with other related parasites as *T. gondii*. Also, the cut off dilution of 1:50 in IFAT showed to be the most appropriated for *N. caninum* serology in dogs. Therefore, we suggest that *N. caninum* immunodominant antigens, specially the 17 and 29–32 kDa proteins, should be selected markers in serological assays for canine neosporosis.

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1. Introduction

Neospora caninum and *Toxoplasma gondii* are structurally, genetically, and immunologically related coccidian protozoan parasites that infect many warm-blooded vertebrates, although neosporosis and toxoplasmosis are biologically two distinct diseases (Dubey, 2003). *T. gondii* has been recognized as an important pathogen for humans and sheep, particularly during

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pregnancy and for immunocompromised patients, whereas *N. caninum* has been considered a major cause of neonatal mortality and/or abortion in cattle and neuromuscular disorders in dogs (Dubey, 2003). Thus, both neosporosis and toxoplasmosis have been recognized as economically important diseases that have considerable impact on the livestock industry (Liao et al., 2005).

N. caninum infection in dogs are epidemiologically relevant because dogs are the major definitive hosts of the protozoan (McAllister et al., 1998), by shedding oocysts into the environment and thus constituting an important risk factor for the horizontal transmission to other animals, particularly cattle (Paré et al., 1998). Clinical and subclinical infections with *N. caninum* in dogs have been extensively reported worldwide with seroprevalence rates ranging from 0.2 to 54% (Moore, 2005). Clinical canine neosporosis is characterized mainly by neuromuscular disorders, but may be generalized with the involvement of several organs, including the skin (Perlé et al., 2001), and it is more severe in young, congenitally infected pups (Barber and Trees, 1998).

A number of serological tests have been developed for the detection of *N. caninum* specific antibodies, but none of them are generally accepted and up to date there is not a standardized test available to be used. Indirect fluorescent antibody test (IFAT) has been often used as a reference test for the detection of *N. caninum* antibodies in dogs, showing very little cross-reactivity with related protozoan parasites (Dubey et al., 1996; Hemphill and Gottstein, 2000). In order to screen a large number of sera, commercially available or home made enzyme-linked immunosorbent assays (ELISA) have been usually employed since they are cost effective and less time consuming. However, the criteria used for interpretation of the results, including the cut off titers, are relatively subjective and they are extremely varied in different seroepidemiological and clinical studies (Lasri et al., 2004). In addition, high background absorbance values and cross-reactivity with antibodies against related parasites have been reported depending on the antigen preparation and the polyclonal antisera used in ELISA (Björkman et al., 1997).

N. caninum antigens that are useful for the diagnosis of neosporosis have been identified and characterized in several domestic animals (Bjerkas et al., 1994). Thus, immunoblotting (IB) assays revealed that immune sera from a wide range of animal species exhibited a similar recognition pattern of four immunodominant (17, 29, 30 and 37 kDa) followed by several minor *N. caninum* antigens. Also, cross-reactive antigens between *N.*

caninum and *T. gondii* have been identified by using monoclonal antibodies, such as the heat-shock protein 70 (HSP70), that could be useful to develop vaccines or drugs to control the diseases caused by both parasites (Liao et al., 2005). Inversely, the identification of these cross-reactive antigens might contribute to identify parasite species-specific antigens to improve the serological diagnosis of both protozoan infections.

The aim of the present study was to evaluate comparatively two serological tests (IFAT and ELISA) for the diagnosis of *N. caninum* infection in dogs through the optimization of cut off titers by using two-graph receiver-operating characteristic curve (TG–ROC) curve and the identification of major cross-reactive antigens with *T. gondii* by the use of inhibition ELISA and IB assays.

2. Materials and methods

2.1. Serum samples

A total of 300 serum samples were obtained from dogs of both genders and different ages and breeds, acquired from the Veterinary Hospital of the Federal University of Uberlândia, private veterinary clinics of the city, and from the Zoonosis Control Center of Uberlândia, Brazil, between 2001 and 2004. Positive control sera were obtained from dogs experimentally infected with *N. caninum* (acquired from VMRD[®] Inc., Pullman, WA, USA) or *T. gondii* (Silva et al., 2002a, 2002b), and negative control sera were collected from healthy dogs with negative serology for both parasites by two serological tests (IFAT and ELISA).

2.2. Parasites and antigens

N. caninum tachyzoites (Nc-1 strain) were obtained by infections of bovine monocytes as host cells, cultured in RPMI 1640 medium supplemented with 3% heat-inactivated calf fetal serum in a 5% CO₂ atmosphere at 37 °C and harvested by scraping off the cell monolayer after 2–3 days of infection (Speer et al., 1985). Tachyzoites were purified by forcible extrusion through a 27-gauge needle and centrifuged at 45 × g for 1 min at 4 °C to remove host cell debris. The supernatant was collected and washed twice (720 × g, 10 min, 4 °C) in phosphate-buffered saline (PBS, pH 7.2). *T. gondii* tachyzoites (RH strain) were maintained in Swiss mice by intraperitoneal serial passage at regular 48 h intervals (Mineo et al., 1980). Mouse peritoneal exudates were harvested and washed in PBS as described above.

N. caninum and *T. gondii* whole antigens were prepared according to Camargo (1964). Parasite suspensions were adjusted to 1×10^6 tachyzoites/mL and treated with 1% formaldehyde for 30 min at room temperature. After washing in PBS, parasites were dry-fixed in microscopic slides and stored at -20°C until being used in IFAT.

N. caninum and *T. gondii* soluble antigens were prepared as described by Scott et al. (1987). Parasite suspensions were adjusted to 1×10^8 tachyzoites/mL, treated with protease inhibitors (aprotinin, 10 $\mu\text{g/mL}$; leupeptin, 50 $\mu\text{g/mL}$; phenyl-methylsulfonyl fluoride [PMSF], 1.6 mM) and then lysed by five freeze–thaw (liquid nitrogen and water bath at 37°C) cycles and further by ultrasound (six 60 Hz cycles for 1 min each) on ice. After centrifugation ($10,000 \times g$, 30 min, 4°C), supernatants were collected, the protein concentration was determined (Lowry et al., 1951) and soluble antigen aliquots were stored at -20°C until being used in ELISA.

N. caninum and *T. gondii* tachyzoites (1×10^8 organisms/mL) were lysed in 4% sodium dodecyl sulfate (SDS) sample buffer and then submitted to electrophoresis in a 12% polyacrylamide gel (SDS–PAGE) under non-reducing conditions (Laemmli, 1970). Proteins were electrophoretically transferred to nitrocellulose membranes (Towbin et al., 1979) and stored at 4°C for further use in IB assays.

2.3. Serological assays

2.3.1. IFAT

Indirect fluorescent antibody tests were carried out to detect IgG antibodies to *N. caninum* (IFAT-Nc) and *T. gondii* (IFAT-Tg) as described by Mineo et al. (2001) and Camargo (1964), respectively. Antigen slides were incubated with dog sera diluted 1:25 (IFAT-Nc) or 1:16 (IFAT-Tg) in PBS and then with a fluorescein isothiocyanate-labeled rabbit anti-dog IgG (Fiocruz/Bio-Manguinhos, Rio de Janeiro, Brazil) diluted 1:50 in PBS plus 0.01% Evans blue. Positive and negative control sera were included in each slide. Slides were examined under fluorescence microscopy and only a bright, linear peripheral fluorescence of the tachyzoites was considered positive.

2.3.2. ELISA

Indirect ELISAs were carried out to detect IgG antibodies to *N. caninum* (ELISA-Nc) and *T. gondii* (ELISA-Tg), according to Silva et al. (2002a). Optimization of the reaction was established in preliminary experiments through block titration of the reagents. Microtiter plates were coated with *N. caninum* (20 $\mu\text{g/}$

mL) or *T. gondii* (10 $\mu\text{g/mL}$) soluble antigens and then incubated with dog sera diluted 1:100 (ELISA-Nc) or 1:64 (ELISA-Tg) in PBS containing 0.05% Tween 20 (PBS-T) and 5% skim milk (PBS-TM). Subsequently, peroxidase-labeled rabbit anti-dog IgG (prepared according to Wilson and Nakane, 1978) diluted 1:8000 in PBS-TM was added and the reaction was developed by adding the enzyme substrate consisting of 0.03% H_2O_2 and ortho-phenylenediamine (OPD) at 1 mg/mL in 0.1 M citrate–phosphate buffer (pH 5.5). The optical density (OD) was read at 492 nm in a plate reader (Titertek Multiskan Plus spectrophotometer, Flow Laboratories, USA). Two positive and six negative control sera were included in each plate for determining the reaction cut off that was calculated as the mean OD for negative control sera plus three standard deviations. Antibody titers were expressed as ELISA index (EI) as described by Silva et al. (2002a) as follows: $\text{EI} = \text{OD}_{\text{sample}} / \text{OD}_{\text{cut off}}$ and values of $\text{EI} > 1.2$ were considered positive.

2.3.3. IB

Immunoblotting assays were performed to verify the reactivity to *N. caninum* (IB-Nc) and *T. gondii* (IB-Tg) immunodominant antigens as previously described (Mineo et al., 2001). *N. caninum* and *T. gondii* proteins blotted on nitrocellulose strips were blocked with 5% skim milk in PBS-T for 2 h at room temperature. Strips were incubated overnight at 4°C with dog sera diluted 1:50 in 1% skim milk in PBS-T, and then with peroxidase-rabbit anti-dog IgG diluted 1:6000 for 2 h at room temperature. Alternatively, serum samples of mice experimentally infected with *N. caninum* or *T. gondii* were also tested and revealed with peroxidase-goat anti-mouse IgG (Sigma Chemical Co., MO, USA) diluted 1:1000. The reaction was developed by adding enzyme substrate (Sigma FastTM 3,3'-Diaminobenzidine tablet sets; Sigma Chemical Co.). Bands were analyzed using the Kodak Digital Science 1D image analysis software and plotted against a standard molecular weight curve to determine their relative mobilities (R_f) and apparent molecular masses. Serum samples were considered positive when showing reactivity to at least two out of three (17, 29–32 and 35–37 kDa) immunodominant antigens of *N. caninum* (Bjerkas et al., 1994; Howe et al., 1998) or to *T. gondii* SAG1 (p30) antigen (Silva et al., 2002b).

2.4. TG–ROC

In order to determine optimal cut off titers in IFAT-Nc and ELISA-Nc, sensitivity and specificity of the tests were evaluated considering IB-Nc as the reference

test and the reactivity to at least two out of three (17, 29–32 and 35–37 kDa) immunodominant antigens of *N. caninum* as infection markers. For this purpose, a total of 200 dog sera were selected and serum samples were serially two-fold diluted starting from 1:25 up to the end-point titer (1:400 for IFAT and 1:3200 for ELISA). Then sensitivity and specificity were determined in different cut-off titers for each test by using the two-graph receiver-operating characteristic curve (Greiner et al., 1995).

2.5. ELISA and IB inhibition

To evaluate the degree of cross-reactivity between *N. caninum* and *T. gondii* antigens, inhibition assays were performed as previously described (Pereira et al., 2005). For ELISA inhibition, optimal serum dilutions were established at 1:50 for *N. caninum* and 1:200 for *T. gondii*, and soluble antigens from parasites were used as inhibitor antigens. Serum samples from seropositive dogs to both parasites were pre-adsorbed (v/v) overnight at 4 °C with 10-fold serial concentrations (200–0.0002 µg/mL) of each inhibitor parasite antigen. Sera incubated with diluent without inhibitors were used as positive controls. After adsorption, the inhibitor–serum mixtures were assayed in ELISA-Nc and ELISA-Tg as described above. The results were expressed as percentage of inhibition and calculated as follows: (OD without inhibitor – OD with inhibitor/OD without inhibitor) × 100 as previously described (Puerta et al., 2005).

For IB inhibition, dog sera with IgG antibodies to both parasites were diluted 1:50 and pre-adsorbed (v/v) with 200 µg/mL of each inhibitor antigen for 2 h at 37 °C. As control, it was used sera incubated with diluent only. Preadsorbed sera were centrifuged at 15,000 × g for 10 min at 4 °C and supernatants were incubated with *N. caninum* and *T. gondii* proteins blotted on nitrocellulose strips, following the same protocol described for IB. Immunoblot profiles were analyzed using the Kodak Digital Science 1D image analysis software by calculating band intensity (*Band Int.*) expressed in pixels. The percentage of inhibition was calculated as follows: (*Band Int.* without inhibitor – *Band Int.* with inhibitor/*Band Int.* without inhibitor) × 100.

2.6. Statistical analysis

The agreement between IFAT-Nc and ELISA-Nc was assessed by calculating the proportion of observed agreement (P_o), the proportion of positive (P_{pos}) and negative (P_{neg}) agreement, and *kappa* (κ) coefficient as previously reported (Cicchetti and Feinstein, 1990).

Seropositivity percentages were compared using the chi-square (χ^2) test (GraphPad Prism 4.0, Graphpad Software Inc.). Values of $P < 0.05$ were considered statistically significant.

3. Results

3.1. Comparison between IFAT and ELISA for *N. caninum*

From 300 dog sera initially tested for IgG antibodies to *T. gondii*, 110 (36.7%) were positive in both IFAT-Tg and ELISA-Tg whereas 190 (63.3%) were negative for both tests. In the next step, the presence of IgG antibodies to *N. caninum* was investigated by IFAT-Nc and ELISA-Nc in both groups of *T. gondii*-seropositive and seronegative dogs. From 300 serum samples tested, 105 (35%) were positive in ELISA-Nc, 32 (10.7%) in IFAT-Nc, being 29 (9.7%) positive in both tests (Table 1), resulting in a relatively good agreement ($P_o = 0.74$) between the tests, but with a low *kappa* coefficient ($\kappa = 0.30$). When positive and negative agreement indices were separately analyzed, a poor concordance between IFAT-Nc and ELISA-Nc was found with regards to positive serological results ($P_{pos} = 0.42$), but a good concordance for negative results ($P_{neg} = 0.83$) was observed.

Seropositivity to *N. caninum* was then evaluated in relation to *T. gondii* serology (Fig. 1), showing significantly higher rates of seropositive samples for *N. caninum* in *T. gondii* seropositive than seronegative dogs in both IFAT-Nc (18.2%) and mostly in ELISA-Nc (55.4%) ($P < 0.001$).

3.2. Recognition of immunodominant antigens of *N. caninum* and *T. gondii*

Immunodominant antigens of *N. caninum* and *T. gondii* were evaluated using sera from both mouse and dog experimentally infected with either *N. caninum*

Table 1
Comparison between IFAT and ELISA results for the detection of IgG antibodies to *N. caninum* in dog sera

ELISA	IFAT		Total n (%)
	Positive n (%)	Negative n (%)	
Positive	29 (9.7)	76 (25.3)	105 (35)
Negative	3 (1.0)	192 (64.0)	195 (65)
Total	32 (10.7)	268 (89.3)	300 (100)

Proportion of observed agreement (P_o) = 0.74; proportion of positive agreement (P_{pos}) = 0.42; proportion of negative agreement (P_{neg}) = 0.83; *kappa* (κ) coefficient = 0.30.

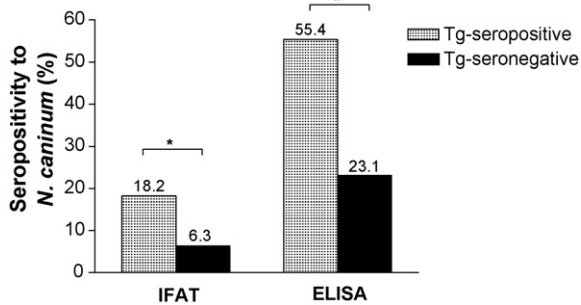


Fig. 1. Serological results obtained by IFAT and ELISA for the detection of IgG antibodies to *Neospora caninum* in *Toxoplasma gondii*-seropositive and -seronegative dogs (* $P < 0.05$).

or *T. gondii* probed in IB (Fig. 2). Both mouse and dog sera infected with *N. caninum* (Fig. 2, lanes 1 and 4, respectively) showed a similar antigenic profile, recognizing at least three immunodominant antigens (17, 29–32 and 35–37 kDa) of *N. caninum*, but *T. gondii* antigens with higher apparent molecular mass (80–85 kDa) were also recognized. Similarly, mouse and dog sera infected with *T. gondii* (Fig. 2, lanes 2 and 5, respectively) recognized at least three of its immunodominant antigens (22, 30 and 43 kDa), but also recognized *N. caninum* antigens above 45–50 kDa. Negative control sera (Fig. 2, lanes 3 and 6) showed weak or no reactivity to minor antigens of both parasites.

3.3. Comparison between IB, IFAT and ELISA for *N. caninum*

Immunoblotting for *N. caninum* (IB-Nc) was performed in a total of 200 dog sera selected on the following criteria: all serum samples with IFAT+/ELISA+ results ($n = 29$), IFAT-/ELISA+ results ($n = 76$), IFAT+/ELISA- results ($n = 3$) and 92 sera randomly selected among the 192 sera with IFAT-/ELISA- results (Table 2). From 29 sera with positive concordant results, 25 (86.2%) recognized at least two out of three *N. caninum* immunodominant antigens whereas only 2 (2.2%) from 92 negative concordant results and 3 (3.9%) from 76 discordant results were considered positive in IB-Nc.

Considering the reactivity for at least two out of three immunodominant antigens of *N. caninum* as a marker of infection, optimal cut off titers in IFAT-Nc and ELISA-Nc were determined by calculating the parameters of sensitivity and specificity of each test using the TG-ROC curve (Fig. 3). The intersection point of the two curves of IFAT (Fig. 3A) and ELISA (Fig. 3B) indicates the cut off point at which equivalent sensitivity and specificity can be achieved. Thus, an optimization of cut off titers was obtained with a titer of 50 for IFAT-Nc and 200 for ELISA-Nc.

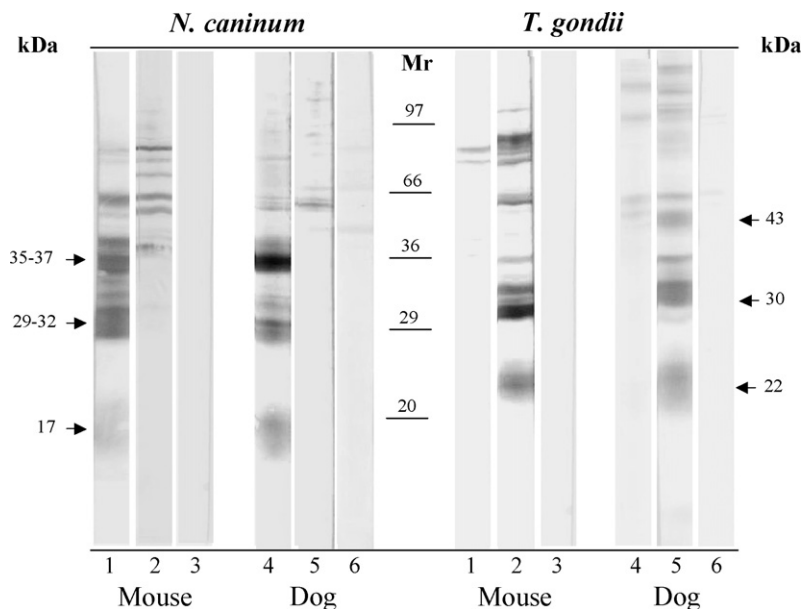


Fig. 2. Immunoblotting of *Neospora caninum* and *Toxoplasma gondii* proteins separated by 12% SDS-PAGE under nonreducing conditions probed with sera from a mouse and a dog experimentally infected with *N. caninum* (lanes 1 and 4) or *T. gondii* (lanes 2 and 5) and respective mouse and dog negative control sera (lanes 3 and 6). Molecular mass markers (M_r) and immunodominant antigens (arrows) of *N. caninum* and *T. gondii* are indicated in kiloDaltons.

Table 2

Reactivity to *Neospora caninum* immunodominant antigens in immunoblotting (IB) of dog sera with concordant or discordant results in IFAT and ELISA for IgG antibodies to *N. caninum*

Results	No. of serum samples	No. (%) of positive sera in IB ^a	Reactivity to <i>N. caninum</i> immunodominant antigens ^b		
			1	2	3
IFAT+/ELISA+	29	25 (86.2%)	4	10	15
IFAT–/ELISA–	92	2 (2.2%)	10	2	0
IFAT–/ELISA+	76	3 (3.9%)	29	3	0
IFAT+/ELISA–	3	0 (0%)	3	0	0
Total	200	30 (15%)	46	15	15

^a Sera were considered positive in IB when at least 2 from 3 immunodominant antigens were recognized.

^b Number of sera reactive to at least 1, 2 or 3 immunodominant antigens (17, 29–32 and 35–37 kDa) of *N. caninum*.

3.4. ELISA and IB inhibition

The degree of cross-reactivity between *N. caninum* and *T. gondii* antigens in dog sera was evaluated

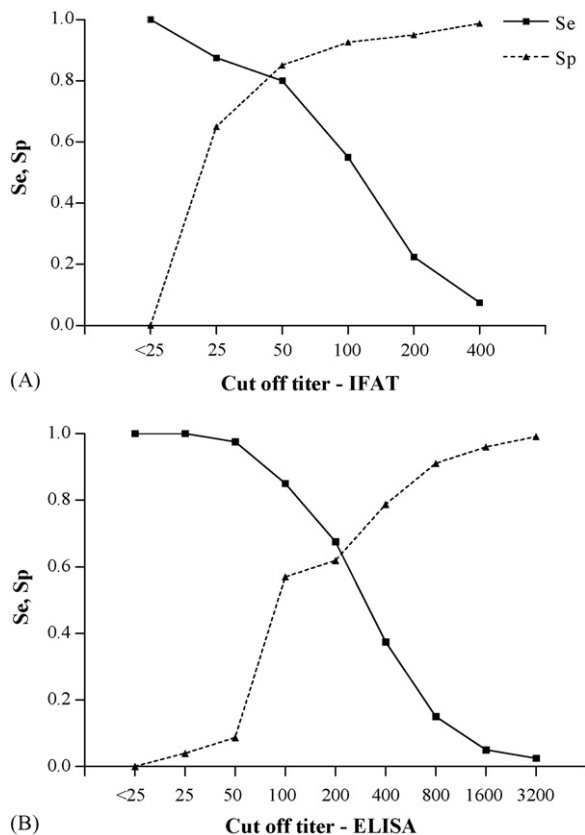


Fig. 3. TG-ROC curve of IFAT (A) and ELISA (B) to detect IgG antibodies to *Neospora caninum* in dog sera. Sensitivity (Se) and Specificity (Sp) of IFAT and ELISA were determined for each cut off titer, considering the reactivity to at least two out of three (17, 29–32 and 35–37 kDa) immunodominant antigens of *N. caninum* as marker of infection. The intersection point of the two curves (Se, Sp) of each test indicates the cut off point at which equivalent sensitivity and specificity can be achieved.

using ELISA inhibition (Fig. 4). ELISA-Nc inhibition curves showed a clear dose-dependent homologous (*N. caninum*) inhibition but also a partial heterologous (*T. gondii*) inhibition (Fig. 4A). In contrast, ELISA-Tg

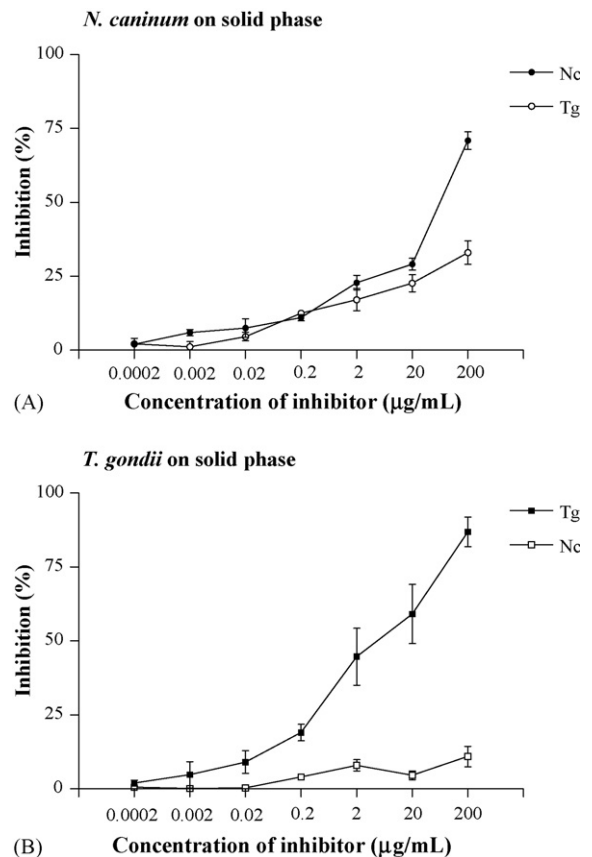


Fig. 4. ELISA inhibition curves with *Neospora caninum* (A) and *Toxoplasma gondii* (B) on solid phase. Dog sera with IgG antibodies to both parasites were pre-adsorbed with different concentrations of *N. caninum* (Nc) or *T. gondii* (Tg) inhibitor antigens. Data represent the mean plus 95% confidence interval of six dog sera with IgG antibodies to both parasites.

inhibition curves showed only homologous (*T. gondii*) dose-dependent inhibition with minor heterologous (*N. caninum*) inhibition (Fig. 4B). In addition, when positive sera to both *T. gondii* and *N. caninum* (Tg+Nc+) and positive to *N. caninum* only (Tg–Nc+) were individually assayed by ELISA–Nc inhibition assay, at maximal concentration (200 µg/mL) of inhibitor antigens, heterologous inhibition was significantly higher in Tg+Nc+ (25%) than Tg–Nc+ (5%) serum samples ($P < 0.05$), indicating some degree of cross-reactivity between *N. caninum* and *T. gondii* antigens (data not shown).

IB inhibition assays were performed to identify which cross-reactive antigens of *N. caninum* and *T. gondii* were recognized by serum samples containing IgG antibodies to both parasites (Figs. 5 and 6). A broad spectrum of *N. caninum* proteins ranging from 17 to 120 kDa were recognized by serum samples in the absence of inhibitor antigens (PBS) and showed a high percentage of inhibition with the homologous antigen, but only proteins of apparent molecular masses above 45–50 kDa showed some degree of *T. gondii* heterologous inhibition (Figs. 5A and 6A). In the same way, a high percentage of homologous inhibition was verified for the majority of proteins of *T. gondii* ranging from 22 to 116 kDa, and

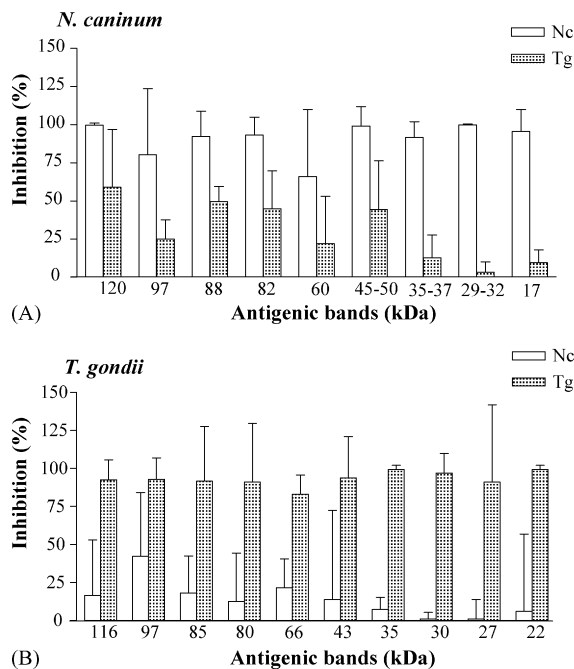


Fig. 5. Inhibition (%) of antigenic bands of *Neospora caninum* (A) and *Toxoplasma gondii* (B) recognized by dog sera when pretreated with *N. caninum* (Nc) or *T. gondii* (Tg) inhibitor antigens at maximal concentration of 200 µg/mL. Data represent the mean plus 95% confidence interval of six dog sera with IgG antibodies to both parasites.

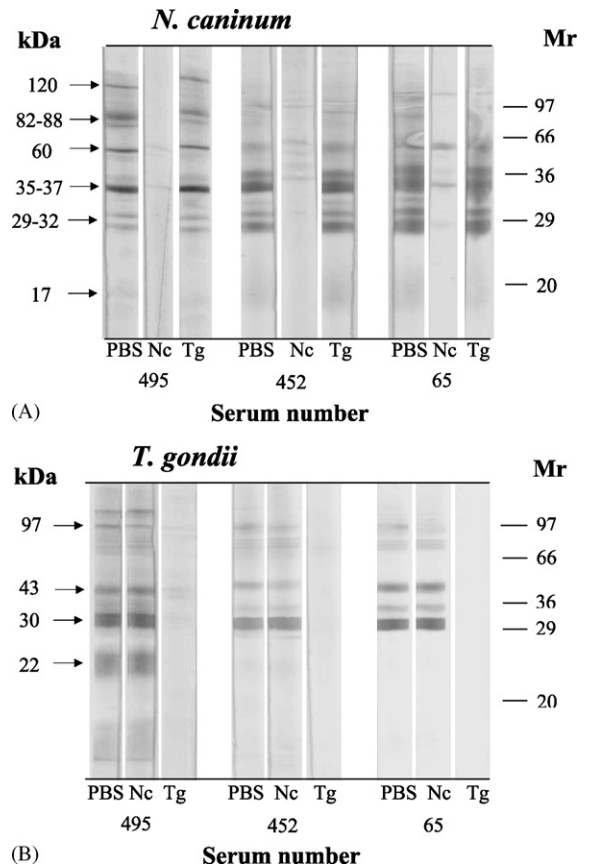


Fig. 6. Immunoblot inhibition of *Neospora caninum* (A) and *Toxoplasma gondii* (B) antigens recognized by three representative sera from dogs with IgG antibodies to both parasites. Sera were pretreated with *N. caninum* (Nc) or *T. gondii* (Tg) inhibitor antigens or with diluent only (PBS). Molecular mass markers (M_r) and immunodominant antigens (arrows) of *N. caninum* and *T. gondii* are indicated in kiloDaltons.

only antigens with apparent molecular masses above 43 kDa showed some degree of *N. caninum* heterologous inhibition (Figs. 5B and 6B). Immunodominant antigens from both parasites (below 35 kDa) showed very low levels of heterologous inhibition, indicating that they were species-specific antigens.

4. Discussion

Diagnosis of *N. caninum* infection in dogs is mainly based on serological assays that have been used as important tools for both clinical and seroepidemiological investigations. However, the criteria used for serum samples to be considered positive are quite variable and depend on the investigator and on the kind of the assays performed. Different dilutions have been considered as positive in different tests, with IFAT titers

ranging from 1:16 to 1:100 (Lasri et al., 2004; Pinheiro et al., 2005; Capelli et al., 2006) and ELISA titers ranging from 1:50 to 1:100 (Lasri et al., 2004; Pinheiro et al., 2005). So far, it has not been possible to make suitable comparisons among prevalence of antibody to *N. caninum* in dogs from these various studies because of the differences in sensitivity and specificity of the employed serological tests and the different cut offs established for each test (Björkman and Ugglä, 1999). Therefore, no conclusive data have been presented to demonstrate which titers might really be considered positive to *N. caninum* among various assays. In addition, low titers of antibodies to *N. caninum* can be the expression of cross-reactivity to related parasites, particularly *T. gondii*, due to common tachyzoite and bradizoyte antigens (Bjerkas et al., 1994; McAllister et al., 1996).

In the present study, we initially compared IFAT and ELISA results for *N. caninum* in dog sera, showing 74% agreement index between the tests, but with a low ($\kappa = 0.30$) kappa coefficient, indicating a poor concordance between these tests for *N. caninum* serology. However, when separate indices of agreement were analyzed, a good association was found regarding the negative agreement index ($P_{\text{neg}} = 0.83$) in contrast to the positive agreement ($P_{\text{pos}} = 0.42$) index. These results were similar to those reported by Lasri et al. (2004) who found a poor positive but good negative agreement between IFAT and ELISA for the serodiagnosis of *N. caninum* infection in dogs.

It is widely known that soluble antigen extracts used in ELISA contain a large number of antigens, particularly those from intracellular origin, whereas parasite surface membrane antigens are preferentially recognized in IFAT. Thus, IFAT has been considered a more species-specific test, although its sensitivity and specificity can be altered based on the established cut off values (Björkman and Ugglä, 1999). In canine neosporosis, the first described ELISA was an iscom ELISA (Björkman et al., 1994) that showed a high concordance with IFAT as the reference test, presenting 98% sensitivity and 96% specificity. Since then, several ELISAs using crude tachyzoite antigens have been developed and tested for *N. caninum* serology in dogs (Lasri et al., 2004; Pinheiro et al., 2005; Capelli et al., 2006), including a commercially available competitive ELISA (cELISA) (VMRD[®] Inc., Pullman, WA, USA), which was recently validated for canine neosporosis using IFAT as the reference test at cut off dilution of 1:16 (Capelli et al., 2006). Therefore, the great variation in cut off titers used for IFAT in different studies could be influencing on the performance of ELISA. For that

reason, ELISA results for *N. caninum* should be interpreted with carefulness and, thus, additional or complementary tests, such as immunoblotting, should be performed in order to clarify the discordant results found by IFAT-Nc and ELISA-Nc.

In this context, we evaluated antigens of *N. caninum* that were recognized by sera from experimentally infected dogs with the parasite by immunoblotting, showing a strong recognition of three immunodominant antigens (17, 29–32 and 35–57 kDa). Interestingly, the great majority (86%) of dog sera with positive concordant (IFAT+/ELISA+) results recognized at least two out of these three antigens, particularly the 29–32 and 35–37 kDa bands, whereas sera with negative concordant (IFAT–/ELISA–) or discordant (IFAT–/ELISA+ or IFAT+/ELISA–) results recognized just one out of these three antigens, especially the 35–37 kDa bands. Recently, Pinheiro et al. (2005) reported that the 35–37 kDa bands of *N. caninum* were recognized by both positive and negative serum samples, indicating that other nonspecific antigens might be present in both groups of dogs. Thus, optimization of cut off titers in IFAT-Nc and ELISA-Nc were determined by TG-ROC curve, considering the reactivity to at least two out of three immunodominant antigens of *N. caninum* as markers of infection, obtaining a titer of 50 for IFAT-Nc and 200 for ELISA-Nc. For IFAT, dilution of 1:50 has been recommended and commonly used for canine sera in several studies (Azevedo et al., 2005; Pinheiro et al., 2005; Andreotti et al., 2006). However, the equivalent sensitivity and specificity values found for ELISA-Nc were lower than those obtained in IFAT-Nc, suggesting that the diversity of crude antigen preparations used in ELISA as well as the affinity characteristics of the secondary antibody used as conjugates might be influencing the performance of the test. A recent study comparing different methods for the solubilisation of *N. caninum* antigens showed that extraction with SDS and sulphobetaines gave higher yields and the best signal versus noise ratio when used in ELISA as compared to the standard sonication method (Zintl et al., 2006). Therefore, the optimal extraction method for parasite proteins depends on the intended application. In this context, Capelli et al. (2006) recently validated cELISA for canine neosporosis, suggesting its use to confirm clinical suspicion of the disease due to its good specificity (96%), but not indicated for seroepidemiological surveys because of the low sensitivity (47%). Thus, the use of this cELISA in canine population requires some techniques for adjustment of misclassification in prevalence and risk-factor studies (Capelli et al., 2006).

When seropositivity to *N. caninum* was evaluated in relation to *T. gondii* serology, a significantly higher number of *N. caninum*-seropositive dogs were found in *T. gondii*-seropositive dogs in both ELISA-Nc and IFAT-Nc tests, particularly for ELISA-Nc. Thus, the degree of cross-reactivity between both parasites was investigated by using inhibition assays, showing a partial heterologous inhibition curve in ELISA-Nc, indicating that antibodies to *N. caninum* were partially adsorbed with *T. gondii* antigens, reflecting in lower reactivity and consequently higher inhibition rates in ELISA-Nc. Such fact was particularly evident in *T. gondii*-seropositive samples, indicating that antibodies to *T. gondii* were also adsorbed with their specific antigens and as a result, lower reactivity and higher inhibition rates also occurred. These findings indicate that the ELISA-Nc here developed showed some degree of cross-reactivity between *N. caninum* and *T. gondii* antigens. Interestingly, inhibition curves for ELISA-Tg showed a clear homologous inhibition, but a minor heterologous inhibition, suggesting that antibodies to *T. gondii* were less adsorbed with *N. caninum* antigens, resulting in higher reactivity and lower inhibition rates in ELISA-Tg. A probable explanation for these findings could be the differential titers of antibodies to *N. caninum* and *T. gondii* in dog sera, with higher titers being found for the former and lower titers for the latter, or possible differences in the antibody affinity for the different antigens.

To identify which cross-reactive antigens of *N. caninum* and *T. gondii* were recognized by dog sera, immunoblot inhibition assays were performed by determining the band intensity in different conditions, with or without inhibitor antigens. A high homologous inhibition was seen for the large spectrum of antigenic bands of both parasites, but heterologous inhibition was more evident for proteins of higher molecular masses, above 45–50 kDa for *N. caninum* and 66 kDa for *T. gondii*. Accordingly, cross-reaction of *N. caninum* and *T. gondii* antigens has been previously reported in hyperimmune rabbit sera (Bjerkas et al., 1994), showing that *N. caninum* protein bands of 46, 88 and 97 kDa were labeled by both anti-*Neospora* and anti-*Toxoplasma* sera and likewise, *T. gondii* proteins of 45, 66 and 85 kDa were recognized by both types of sera. A recent study has reported that several proteins with apparent molecular masses around 50–60 kDa were recognized by *N. caninum*- and *T. gondii*-negative dog sera, suggesting nonspecific recognition to unrelated organisms (Pinheiro et al., 2005). Also, cross-reactive antigens between *N. caninum* and *T. gondii* were recently identified and characterized by monoclonal

antibodies produced against *N. caninum* (Liao et al., 2005), and consisted of proteins located on the surface of parasites (molecular masses from 28 to 76 kDa), on interior organelles (molecular masses of 35, 50 and 64 kDa) and on the apical end of parasites (unknown molecular masses). Three of these proteins could be identified as the heat shock protein 70 (HSP70), the protein disulfide isomerase (PDI), and ribosomal protein 1 (RP1) (Liao et al., 2005). Proteomic analyses in two-dimensional electrophoresis (2-DE) immunoblotting using rabbit antiserum against *N. caninum* have also identified a number of homologue proteins, ranging from 41 to 76 kDa, between *N. caninum* and *T. gondii* tachyzoites, such as HSP70, PDI, tubulin α - and β -chains, actin and enolase, which were believed as common antigens between both parasites (Lee et al., 2005).

On the other hand, in our study, immunodominant antigens from both parasites, especially those below 35 kDa, were slightly inhibited by heterologous antigens, indicating that they are more species-specific antigens. Accordingly, Bjerkas et al. (1994) found that anti-*Toxoplasma* rabbit sera exhibited faint and inconsistent binding to the immunodominant (17, 29, and 35–37 kDa) *N. caninum* antigens and that proteins with apparent molecular masses between 17 and 37 kDa were recognized by sera from infected animals of several species, including dogs. On the 2-DE maps, proteins ranging from 26–33 kDa (NcGRA2) and from 31–40 kDa (NcGRA7), two proteins of *N. caninum* dense granules, exhibited low homology with the respective *T. gondii* dense granule proteins, suggesting as possible species-specific proteins of *N. caninum* (Lee et al., 2005).

Altogether, the results of our study indicate that ELISA should be used critically when crude tachyzoite antigen preparations are employed, due to possible cross-reactivity with other related parasites as *T. gondii*. In addition, the cut off dilution of 1:50 in IFAT showed to be the most appropriated for *N. caninum* serology in dogs. Therefore, we suggest that *N. caninum* immunodominant antigens, specially the 17 and 29–32 kDa proteins, should be selected markers in order to develop serological diagnostic systems for *N. caninum* infection in dogs.

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